



Started on	Thursday, 6 August 2026, 11:36 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:38 AM
Time taken	2 mins 16 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d",&n);
6
7      int arr[n];
8
9      for(int i=0;i<n;i++)
10     {
11         scanf("%d",&arr[i]);
12     }
13
14     int count=0;
15
16     for(int i=0;i<n;i++)
17     {
18         if(arr[i]==0){
19             count=count+1;
20         }
21     }
22
23     printf("%d",count);
24 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1 1	0	0	✓

	Input	Expected	Got	
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Thursday, 6 August 2026, 11:38 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:17 AM
Time taken	20 days 23 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int majorityElement(int nums[], int n) {
4      int candidate = 0;
5      int count = 0;
6      for (int i = 0; i < n; i++) {
7          if (count == 0) {
8              candidate = nums[i];
9          }
10         if (nums[i] == candidate)
11             count++;
12         else
13             count--;
14     }
15     return candidate;
16 }
17
18 int main() {
19     int n;
20     scanf("%d", &n);
21
22     int nums[n];
23
24     for (int i = 0; i < n; i++) {
25         scanf("%d", &nums[i]);
26     }
27     printf("%d\n", majorityElement(nums, n));
28     return 0;
29 }
30

```

	Input	Expected	Got	
✓	3	3	3	✓
	3 2 3			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

**Thaneeshga M 251801354-2025-AIDS** ▾**T2****Started on** Thursday, 6 August 2026, 11:39 AM**State** Finished**Completed on** Thursday, 27 August 2026, 11:21 AM**Time taken** 20 days 23 hours**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, x, l = 0, r, ans = -1;
5      scanf("%d", &n);
6
7      int a[n];
8      for(int i = 0; i < n; i++)
9          scanf("%d", &a[i]);
10
11     scanf("%d", &x);
12     r = n - 1;
13
14     while(l <= r) {
15         int m = (l + r) / 2;
16
17         if(a[m] <= x) {
18             ans = a[m];
19             l = m + 1;
20         } else
21             r = m - 1;
22     }
23
24     printf("%d", ans);
25     return 0;
26 }
27

```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓

	Input	Expected	Got	
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

**Thaneeshga M 251801354-2025-AIDS** ▾**T2****Started on** Thursday, 27 August 2026, 11:21 AM**State** Finished**Completed on** Thursday, 27 August 2026, 11:22 AM**Time taken** 49 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int f(int a[], int l, int r, int x) {
4      if (l >= r) return 0;
5      if (a[l] + a[r] == x) {
6          printf("%d\n%d", a[l], a[r]);
7          return 1;
8      }
9      if (a[l] + a[r] < x)
10         return f(a, l + 1, r, x);
11     return f(a, l, r - 1, x);
12 }
13
14 int main() {
15     int n, x;
16     scanf("%d", &n);
17     int a[n];
18
19     for (int i = 0; i < n; i++) scanf("%d", &a[i]);
20     scanf("%d", &x);
21
22     if (!f(a, 0, n - 1, x))
23         printf("No");
24
25     return 0;
26 }
27

```

	Input	Expected	Got	
✓	4 2 4 8 10 14	4 10	4 10	✓
✓	5 2 4 6 8 10 100	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Thursday, 27 August 2026, 11:22 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:23 AM
Time taken	54 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int cmp(const void *a, const void *b) {
5      return (*(int *)a - *(int *)b);
6  }
7
8  int main() {
9      int n, a[100];
10     scanf("%d", &n);
11
12     for(int i = 0; i < n; i++)
13         scanf("%d", &a[i]);
14
15     qsort(a, n, sizeof(int), cmp);
16
17     for(int i = 0; i < n; i++)
18         printf("%d ", a[i]);
19
20     return 0;
21 }
22

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

